

Statistical and Computational Models

Lecture 1: Statistical and computational foundations

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Department of Government, Cornell University · Fall 2026

Roadmap

The object of estimation

Estimating f

Loss and expected prediction error

What is 'Learning'?

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For example: Y_i is whether i voted, $\mathbf{X}_i$ is what we know about them, and $p(\mathbf{y} | \mathbf{x})$ tells us how the two are connected. We never observe $p(\mathbf{y} | \mathbf{x})$, instead we hope to learn it from a finite sample of pairs.

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Question: is linear regression a form of statistical learning? Yes! We 'learn' the $\hat{\beta}$ that minimises the sum of squared residuals.

Estimand

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With a coherent estimand in place, you can then turn to questions of **estimation**.

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Estimand: the unobserved parameter or function we want.

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Remember: the estimand is fixed and unknown. The estimate changes with every sample.

Some vocabulary

Statistics	Machine learning
estimation	learning, training
covariate, regressor, independent variable	feature
outcome, response, dependent variable	label, target
model selection	tuning, model selection
out-of-sample prediction error	generalization error, test error
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It is worth being familiar with both; we will sometimes switch back and forth.

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$$Y = f(\mathbf{X}) + \varepsilon$$

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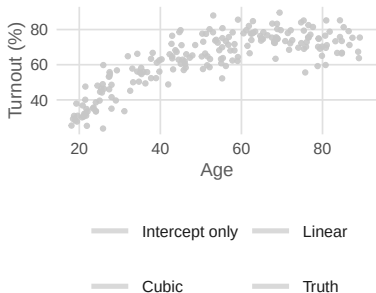
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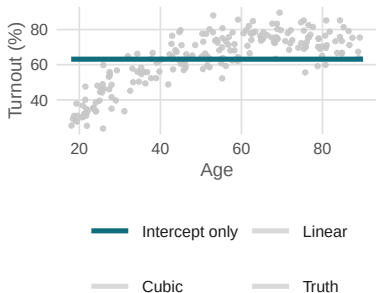
Change $\mathbf{X}$ and you change Y via $f(\cdot)$. This may or may not be causal.

Models as abstractions of reality



Suppose we have 200 age-by-state units (e.g. one might be the turnout rate for 20-year olds in NY).

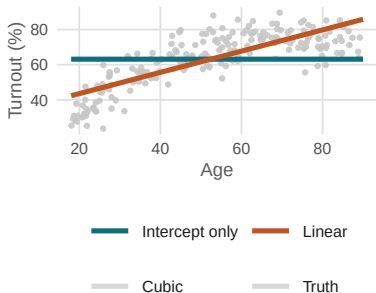
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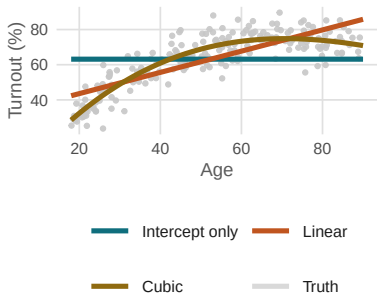


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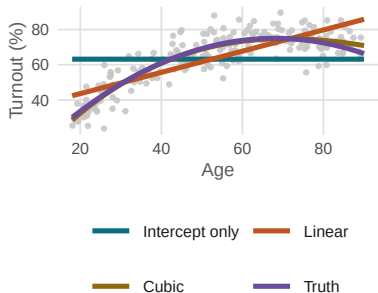
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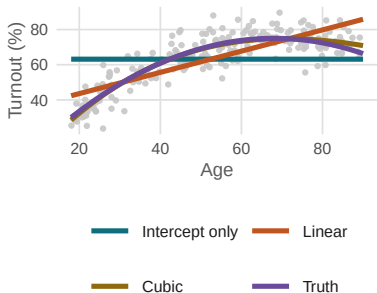
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Each fitted curve is a distinct **estimator** $\hat{f}$, returning its own **estimates** on a given draw.

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Read: the first term is a property of the **estimator** you chose. The second is a property of the **data generating process**.

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Linear	50.85	49	100.34
Cubic	1.34	49	49.73
Truth	0.00	49	48.58

The **irreducible** column never moves: it is σ^2 .

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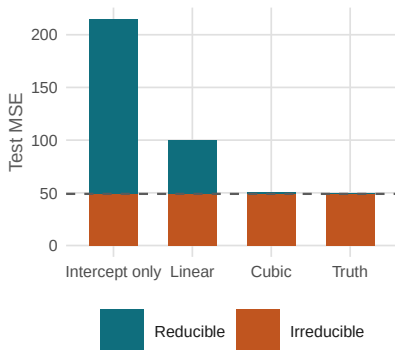
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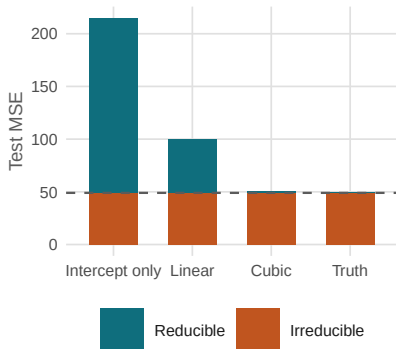
Note: we can only **split** the last column because this is a simulation – we wrote f and σ^2 ourselves. With real data you see **test MSE alone**.

The error floor



The dashed line is $\sigma^2 = 49$ – seven points of turnout, squared.

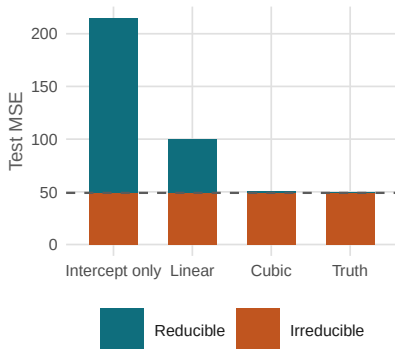
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The **irreducible** block is the same height everywhere.

Roadmap

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Estimating f

Loss and expected prediction error

Two routes to $\hat{f}$

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Assume a form: $f(\mathbf{x}) = \beta_0 + \beta_1 \mathbf{x}$.

Estimate $p + 1$ parameters.

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We will spend some time over the next two weeks thinking about what you trade off when choosing between the two.

Supervised and unsupervised

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'Supervised' because the outcome constrains the answer. It **teaches**.

Continuous Y is **regression**.

Categorical Y is **classification**.

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Most of the course is **supervised** – in social science we usually observe Y – but we will take a few detours into the **unsupervised** world.

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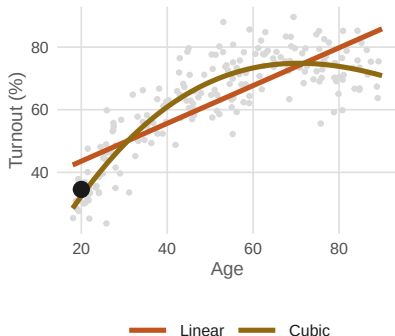
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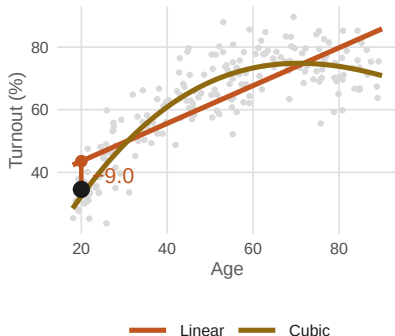
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Holding one cell out



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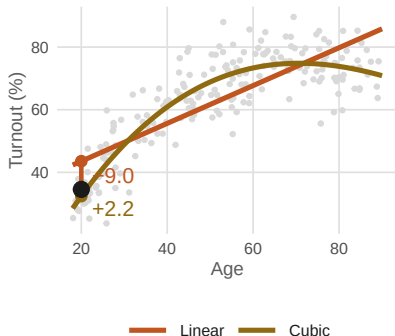
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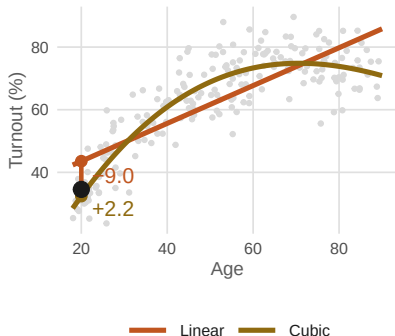


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For two different models the same unit gives us two different **errors**.

Two predictions on one case

Predictor	Age	Turnout	Predicted	Raw	Squared	Absolute
Linear	20	34.6	43.6	-9.0	81.0	9.0
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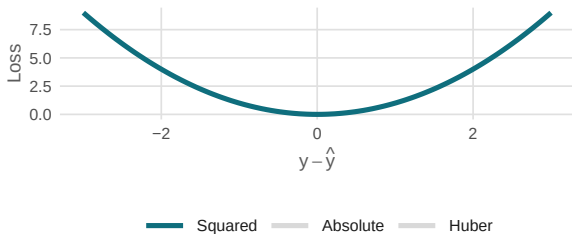
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So the two costs **need not agree**.

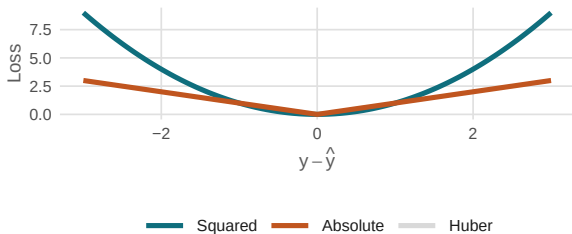
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Squared: $(y - \hat{y})^2$. Double the error, quadruple the cost.

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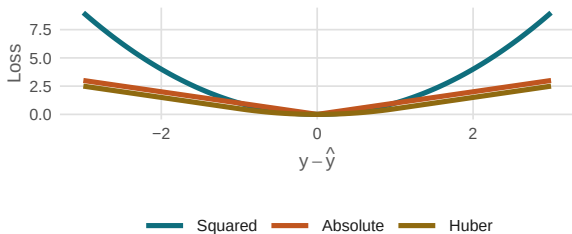


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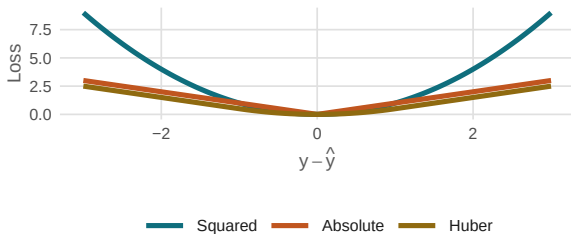
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Read: the loss function **prices the errors we necessarily make**. That choice is substantive, not a **computational convenience**.

Expected prediction error

$$\text{EPE}(f) = \mathbb{E} \left[L(Y, f(\mathbf{X})) \right] = \int L(y, f(\mathbf{x})) p(y, \mathbf{x}) dy d\mathbf{x}$$

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$$f^* = \arg \min_f \text{EPE}(f)$$

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$$\text{EPE}(f) = \mathbb{E}[L(Y, f(\mathbf{X}))] = \int L(y, f(\mathbf{x})) p(y, \mathbf{x}) dy d\mathbf{x}$$

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Read: EPE is the average loss over the **data generating process**. f^* is the function that minimises it.

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Read: EPE is the average loss over the **data generating process**. f^* is the function that minimises it.

Note: ISL calls this the **expected test MSE** – squared loss, at one $\mathbf{x}$. Elsewhere you will meet it as the **risk**.

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Estimating EPE **in sample** causes problems, which we will address soon.

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ISL 2.1 and Spirling & Stewart are the core reading for today.

Lundberg et al. arrives as core reading at lecture 4. The rest are optional, and sit under Unit 1 on the course website.